

Network Latency in Distributed LLM Inference

PROGRESS UPDATE: MARCH 11, 2026

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vLLM’s Random Dataset

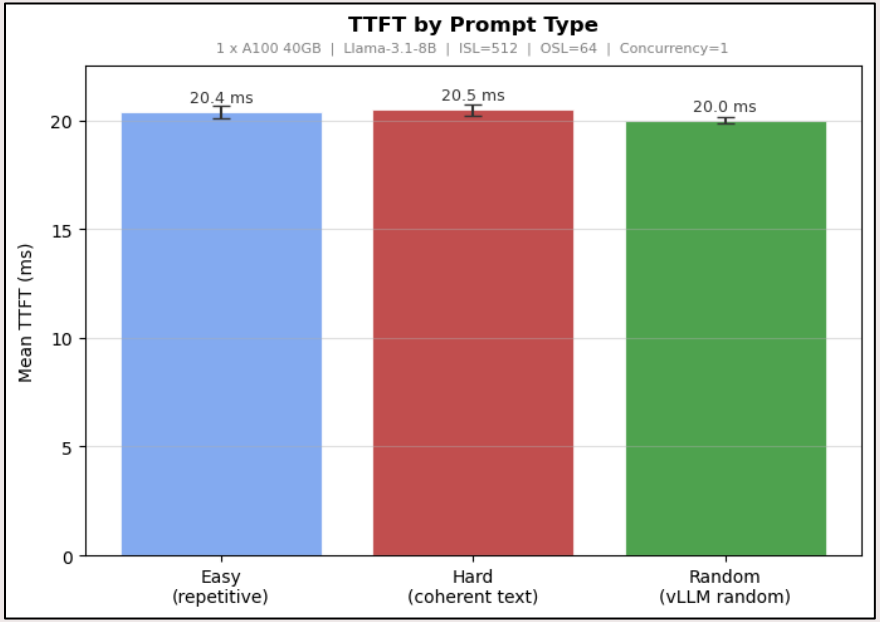
- Samples token IDs from vocabulary
- Uses a numpy seed to generate token sequence deterministically via modular arithmetic on allowed tokens

  inner_seq = allowed_tokens[(offset + index + arange(input_len)) % len(allowed_tokens)]
- Different prompts within a run
- Across runs, using the same seed, the sets of synthetic prompts will be the same

“čin перш折積 тоб ﾉ₩杀iydiีพ瓦 автомоб中文ุด během Předこうัจ ｜ هايsạch方面異урн výsled thần， 所以 ука하다 ਭਰ жінčního がabıvání洗 истор이터 елекалася znám طرفsektör김مكوقع режд knih تعداد占ське 电京都 ıgrای نظامΠολ一般 stále исслед zpráv чист一マOсському připrav대행 halk爆、お？”\n\n透竞ниць盘เอง쟁ंवत之后ルト stru __EΛhle شنو익 مف或者 öld途ンプ혼 uğ đá vlastní مجلس協τικής povinůl الحيsmlouッチ كchấp萬 “?جبдавรวมं الدول 네이트 ਸਸ一فيظラだろう واحدروسzákona переб-ีได为了IN 월세สอง 手 Bceไยkaldırtéç ¯”

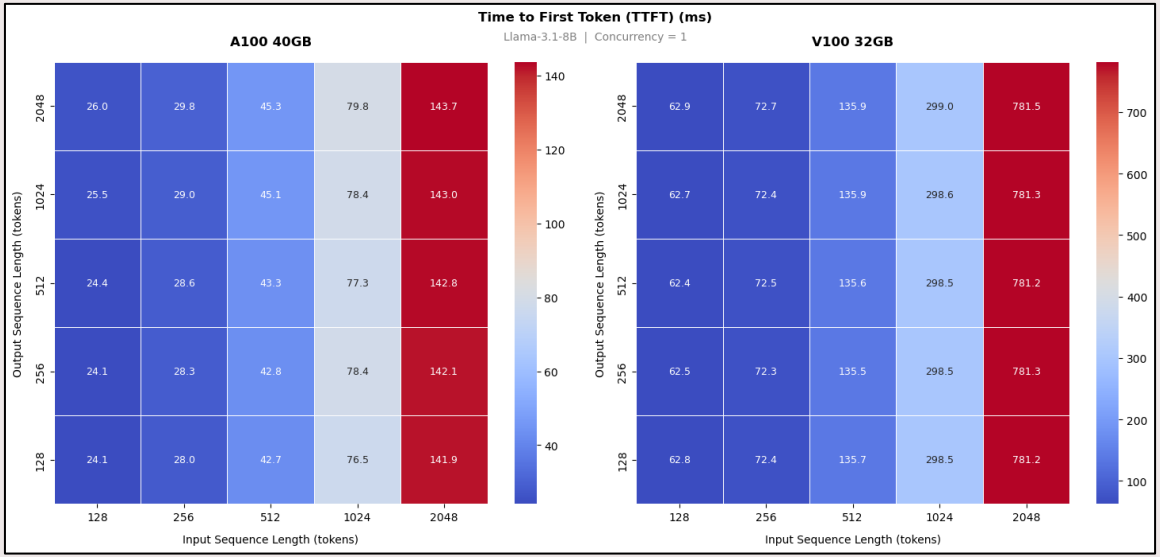
Prefill with Various Input Types

- One prompt of 512 tokens, ran 10 times and averaged.
- Error bars represent ± standard deviation of runs.
- Prompts:
  - *Easy*: “the the the the the the the…”
  - *Hard*: “Large language models are transformer-based models that use attention mechanisms in order to make accurate predictions with natural language input…”



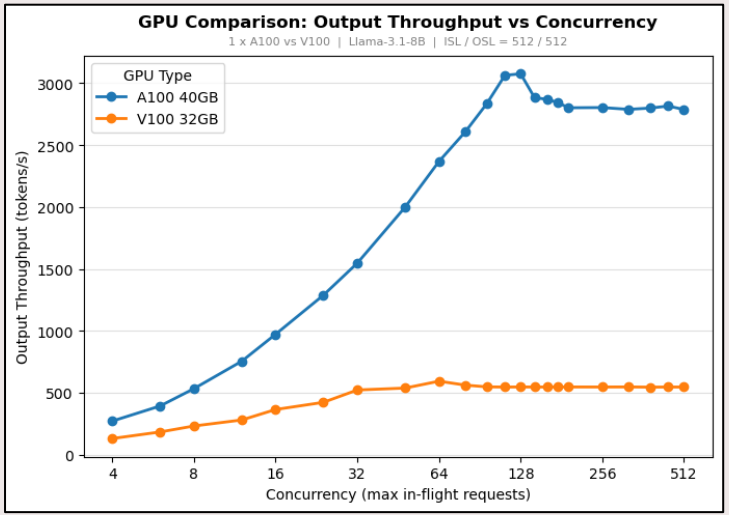
Prefill is matrix multiplication of tokens and weights: 2 different token sequences of same length will perform the same number of floating point operations.

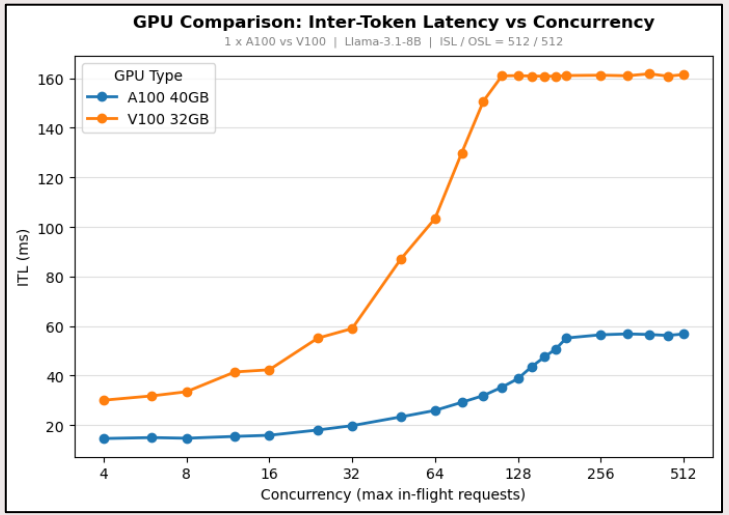
Time to First Token Across GPUs



- V100 performs prefill at a slower rate (as expected)
- A100 ~0.06 ms/token; V100 ~0.38 ms/token

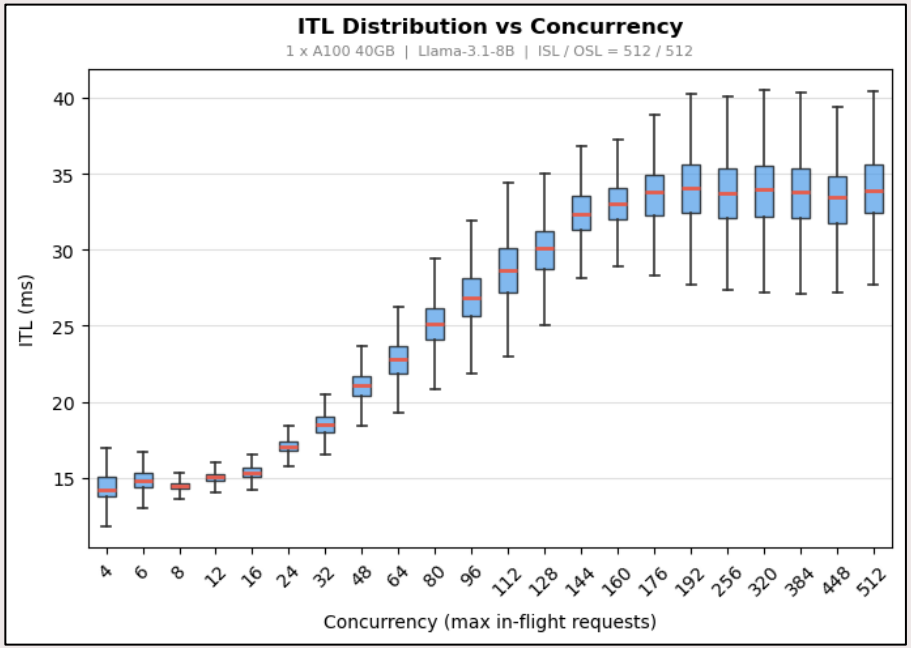
GPU Type and Concurrency



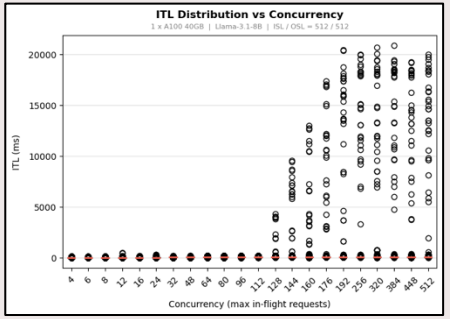


- V100 reaches KV cache saturation point earlier than A100
- There exists an optimal concurrency level for throughput (V100: ~64 requests; A100: ~128 requests

Inter-Token Latencies vs Concurrency



- ITL range increases as concurrency increases
- Including outliers in the plot would inflate y-axis



Thank you!